Statistical Learning and Data Analysis Module B

Resampling methods for Model Assessment and Selection

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A thought

«It's difficult to make predictions, especially about the future" –Niels Bohr

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Key goals in Supervised Learning

Model Selection - the process of selecting the proper level of flexibility for a class of models: *estimating the performance of different models in order to choose the best one.*

Model Assessment - the process of evaluating the performance of a specific model: *having chosen a final model, estimating its prediction error (generalization error) on new data.*

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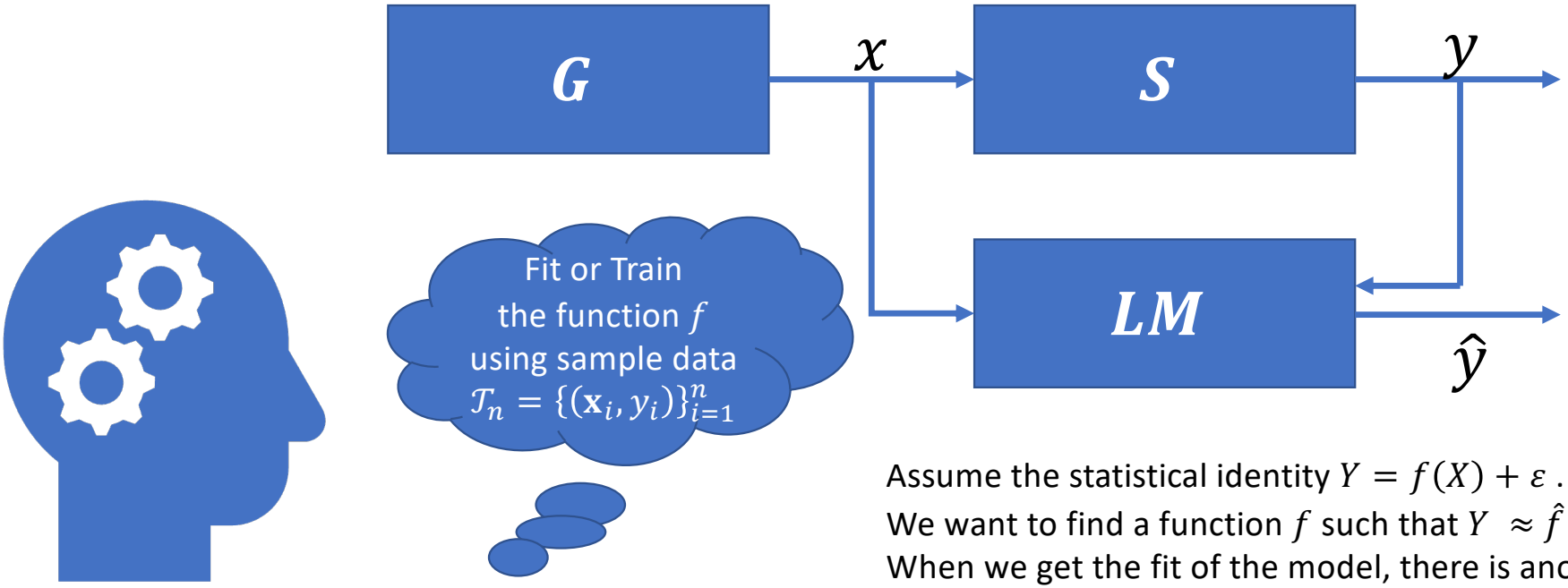

How do we estimate f?

Assume the statistical identity $Y = f(X) + \varepsilon$.
We want to find a function f such that $Y \approx \hat{f}(X)$.
When we get the fit of the model, there is another identity:

$$Y = \hat{f}(X) + e$$

where $\hat{f}(X) = \hat{Y}$ is the prediction model and $e = Y - \hat{f}(X)$ is the residual, estimation of the error ε (prediction error).

Let $(Y - \hat{Y})$ be the prediction error. The *expected value* of the squared difference between the actual value and the predicted value of Y can be decomposed as:

$$E(Y - \hat{Y})^2 = E[f(X) + \varepsilon - \hat{f}(X)]^2 = E[\hat{f}(X) - f(X)]^2 + var(\varepsilon) = var\left(\hat{f}(X)\right) + \left[bias\left(\hat{f}(X)\right)\right]^2 + var(\varepsilon)$$

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Regression Problem

- Output (target) numerical Y
- A vector of inputs (predictors) X
- A prediction model $\hat{f}(X)$ that has been estimated from a training set $\mathcal{T}$
- The loss function $\mathcal{L}(Y, \hat{f}(X))$ for measuring errors (residual) between Y and $\hat{f}(X)$.
- Typical choices of loss function are:

$$\mathcal{L}(Y, \hat{f}(X) = \begin{cases} \left(Y - \hat{f}(X)\right)^2 \\ \left|Y - \hat{f}(X)\right| \end{cases}$$

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Test error *versus* Training error

- **Test error**, also refereed as **generalization error**, is *the prediction error over an independent test sample*

$$Err_{\mathcal{T}} = E[\mathcal{L}(Y, \hat{f}(X)|\mathcal{T}]$$

- where both X and Y are drawn randomly from their joint distribution (population). Here *the training set is fixed, and test error refers to the error for this specific training set*.
- The ***Expected test error*** (or **Expected prediction error**) averages over everything that is random, including the randomness in the training set that produced $\hat{f}$:

$$Err = E\left[\mathcal{L}(Y, \hat{f}(X)\right] = E[Err_{\mathcal{T}}]$$

- **Training error** is *the average loss over the training sample*

$$\overline{err} = \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}(y_i, \hat{f}(x_i))$$

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Bias, Variance and Model Complexity

We would like to know the expected test error of our estimated model $\hat{f}$.

- As the model becomes more and more complex, it uses the training data more and is able to adapt to more complicated underlying structures. Hence there is a decrease in bias but an increase in variance.
- There is some intermediate model complexity that gives minimum expected test error.
- Unfortunately training error is not a good estimate of the test error. Training error consistently decreases with model complexity, typically dropping to zero if we increase the model complexity enough. However, a model with zero training error is overfit to the training data and will typically generalize poorly.

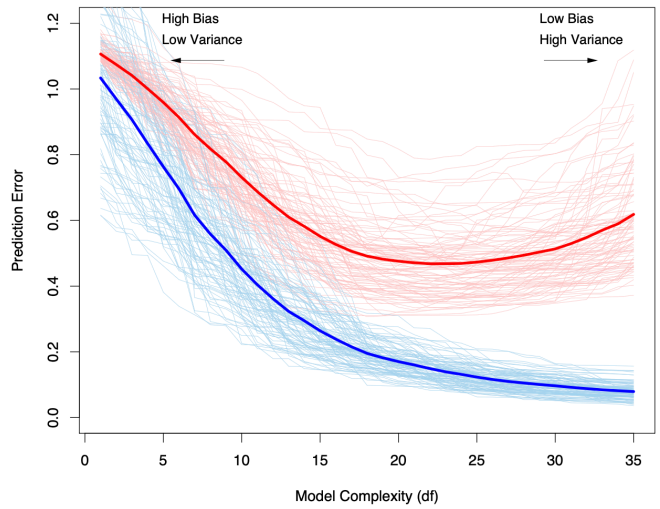

FIGURE 7.1. *Behavior of test sample and training sample error as the model complexity is varied. The light blue curves show the training error* $\overline{\mathrm{err}}$, *while the light red curves show the conditional test error* $\mathrm{Err}_{\mathcal{T}}$ *for* 100 *training sets of size* 50 *each, as the model complexity is increased. The solid curves show the expected test error* Err *and the expected training error* $\mathrm{E}[\overline{\mathrm{err}}]$.

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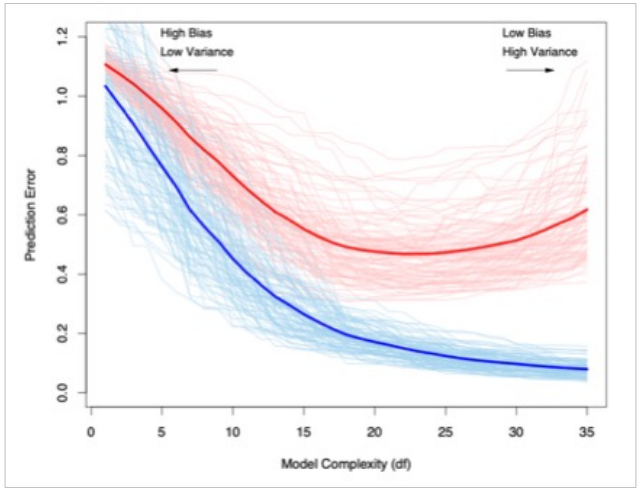
Test and training error as a function of model complexity

- This is the typical behavior of the test and training error, as model complexity of the prediction model $\hat{f}(X)$ is varied.
- The training error tends to decrease whenever we increase the model complexity, that is, whenever we fit the data harder.
- However, with too much fitting, the model adapts itself too closely to the training data, and will not generalize well (i.e., have large test error). In that case the predictions will have ***large variance***.
- In contrast, if the model is not complex enough, it will underfit and may have ***large bias***, again resulting in poor generalization.
- We will discuss methods for estimating the test error of a statistical model (machine learning) class, hence estimating the optimal amount of model complexity for a given machine learning model class and training set ***(Vapnik–Chervonenkis (VC)-dimensionality theorem).***



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Resampling Methods

Resampling methods are an indispensable tool in modern statistics. They involve ***repeatedly drawing samples from a training set*** and refitting a model of interest on each sample in order to obtain additional information about the fitted model.

Resampling methods were developed when the access to computer became easier and easier. They can be, ***computationally expensive*** because they involve fitting the same statistical method multiple times using different subsets of the training data.

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Two commonly used resampling methods

Cross-Validation can be used to estimate the test error associated with a given statistical learning method in order to evaluate its performance, or to select the appropriate level of flexibility.

Bootstrap is mostly used to provide a measure of accuracy of sample parameter estimates (confidence intervals).

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The Validation Set Approach

- The data set $\mathcal{X}$ is randomly split in two data sets:
  - a learning (or training) set used to train the statistical learning method;
  - a test/validation set (or hold-out set) used to check the goodness of fit through either MSE of error rates for prediction.

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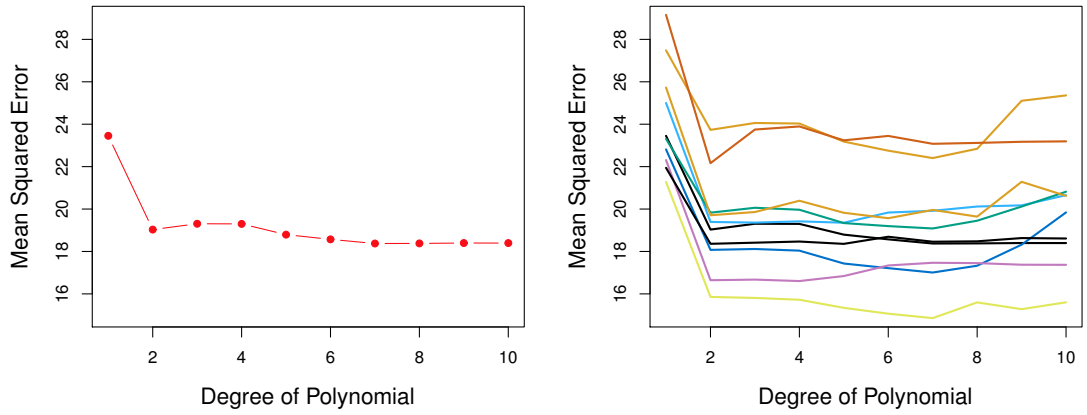

Auto data set: Predicting **Mpg** using polynomial functions of **horsepower**

✓ ***Left:*** Validation MSE estimates for a single split into training and test data sets.

✓ ***Right:*** Validation MSE estimates for ten times repeated split into training and test data sets → High variability in the MSE estimates.

Drawbacks of Validation Set Approach

- The validation estimate of the test error rate can be highly variable, depending on precisely which observations are included in the training set and which observations are included in the validation set.
- Only a subset of the observations are used to fit the model. The validation set error rate may tend to overestimate the test error rate for the model fit on the entire data set.

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Cross-Validation

- Cross-validation (CV) is a resampling method generally used for model selection.
- It is closely related to the test set (or validation set) procedure, with some substantial differences.

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Cross-Validation Procedures

- Cross-validation is a refinement of the validation set procedure.
- Let's give a look to
  - Leave-One-Out-Cross-Validation (LOOCV)
  - Generalized cross validation
  - V –fold cross-validation

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Leave-One-Out-Cross-Validation

- Let $\mathcal{X}$ be a set of data of n observations.
- LOOCV involves splitting the set of observations into two parts (learning and validation): $\mathcal{X}_1$ is the test set and $\mathcal{X}_{2:n}$ is the learning set.
- The statistical learning method is fit on the $(n-1)$ training observations, and a prediction is made for the excluded observation.
- The procedure is repeated $(n-1)$(more) times by excluding from the training set one observation and using it as validation set.
- For each of the $i = 1, \dots, n$ replications, the test set error is computed. Let's call it MSE_i .
- The LOOCV estimates of the MSE is

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} MSE_i$$

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Generalized Cross-Validation

- LOOCV is very computationally intensive to implement since the model has to be fit n times.
- If the statistical learning model is a least squares linear or polynomial regression, the following criterion (called Generalized Cross Validation - GCV) produces the same result of LOOCV but fitting only one single model:

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2$$

where y_i and $\hat{y}_i$ are the observed and fitted values of the response variable and h_{ii} is the diagonal element of the Hat matrix $H = X(X'X)^{-1}X'$.

- CV is really useful when the sample size is quite large. On the other hand, LOOCV can be used with any statistical learning method, being a very general method.

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V-fold Cross-Validation

- This approach involves randomly V-fold CV dividing the set of observations into V groups, or folds, of approximately equal size.
- The first fold is treated as a validation set, and the method is fit on the remaining $(V-1)$ folds. The error, say MSE_i, is computed on the observations in the held-out fold.
- This procedure is repeated V times; each time, a different group of observations is treated as a validation set.
- The V -fold CV error is

$$CV_{(V)} = \frac{1}{V}\sum_{i=1}^{V} MSE_i$$

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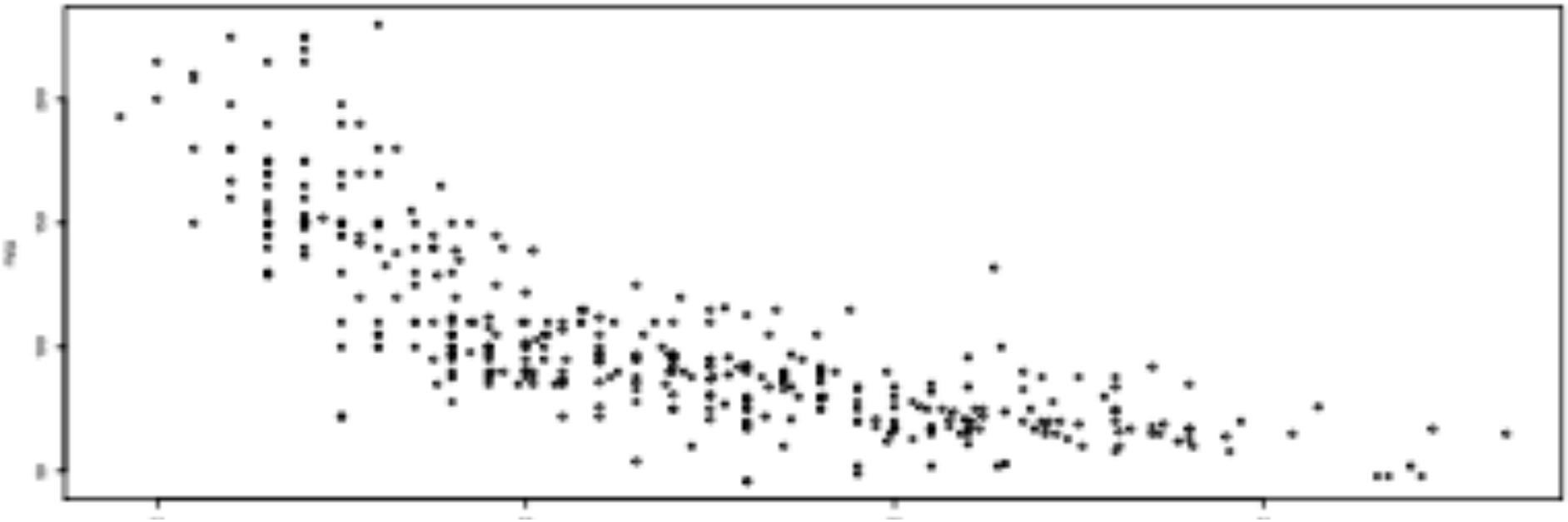
Cross-Validation: An example

- **Auto data set**: Predicting **Mpg** using polynomial functions of **horsepower**



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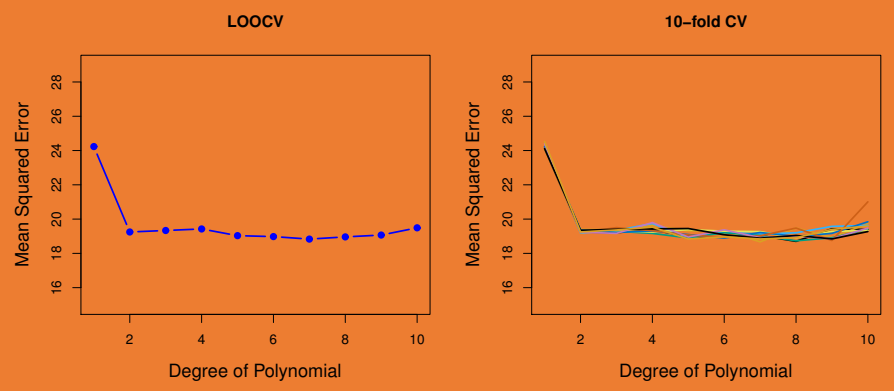
LOOCV *versus* 10-fold CV



10-fold CV was run ten separate times each with a different random split of the data into ten parts

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LOOCV *versus* V-fold CV

- LOOCV is computationally intensive whereas 10-fold CV is more flexible
- Exactly as LOOCV, V -fold CV is a general method that can be applied to a very large variety of statistical learning methods.
- GCV is identical to LOOCV, but it can be used only for linear (or polynomial) least squares problems.
- V −fold CV is a particular case of LOOCV. In the latter case, $V = n$.
- V-fold CV is computationally faster than LOOCV.
- In general, the user choose either $V = 5$ or $V = 10$.

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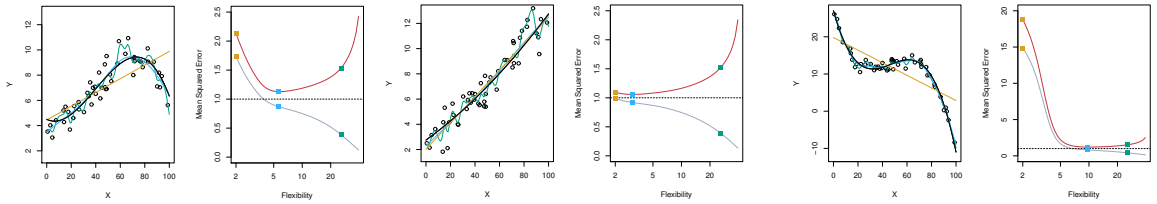
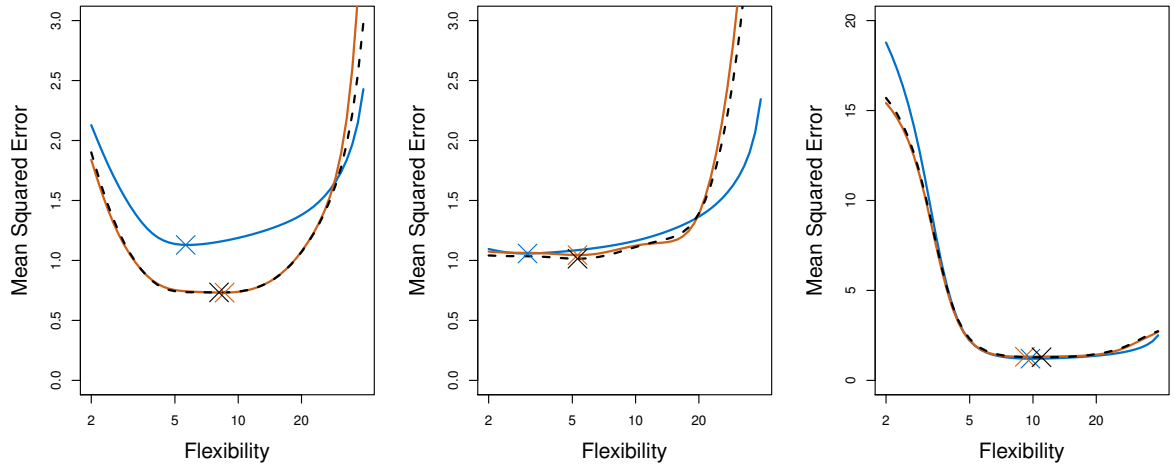
Simulated data sets





- All the CV curves come close to identifying the correct level of flexibility (i.e., the minimum point in the estimated test MSE curve)

✓ **True test MSE: Blue curve**

✓ **LOOCV: Black dashed curve**

✓ **10-fold CV: Orange curve**

➢ *The cross is the minimum point.*

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Bias-Variance Trade-Off

- **LOOCV** is the method that should be preferred if our goal is *to reduce the bias*.
- LOOCV has *larger variance than V-fold CV* because there is much less overlap among the training sets with respect to V-fold CV. *Overlap means correlation*.
- Since the mean of many highly correlated quantities has higher variance than does the mean of many quantities that are not as highly correlated, the test error estimate resulting from LOOCV tends to have higher variance than does the test error estimate resulting from V-fold CV.
- Empirically, choosing either $V = 5$ or $V = 10$ allows test error rate estimates that suffer neither from excessively high bias nor from very high variance.

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Cross-Validation on Classification Problems

- When the response or target Y is qualitative, rather than using the MSE to quantify test error, we use the number of misclassified observations.
- LOOCV error rate can be defined as

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} I(y_i \neq \hat{y}_i)$$

where $I(y_i \neq \hat{y}_i)$ is the indicator function which is equal to 1 if $y_i \neq \hat{y}_i$, 0 otherwise.

- The V-fold CV error rate and validation error rates are defined analogously.

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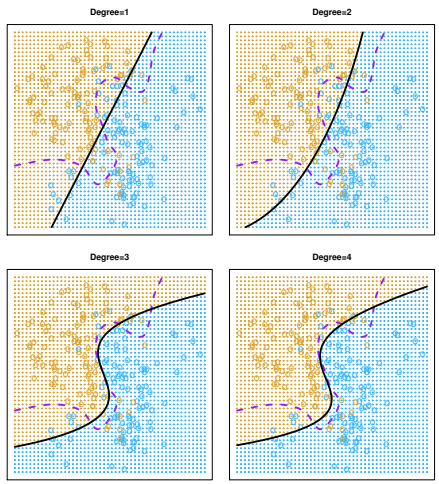
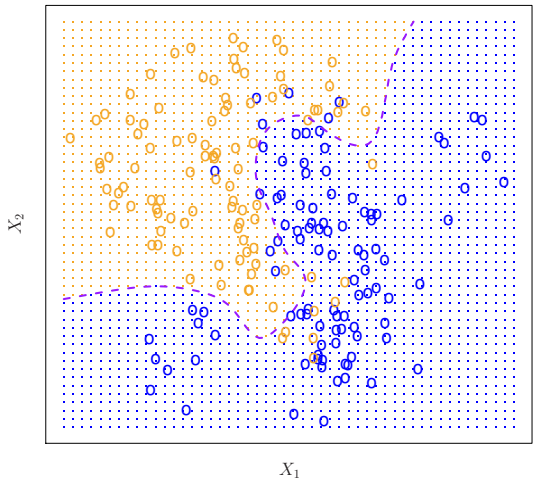
Logistic Regression fitting with non-linear decision boundary

Using polynomial functions of the predictors (degrees 1—4).

The test error rates are respectively 0.201, 0.197, 0.160, 0.162, while the Bayes error rate was

$1 - \mathrm{E}\left(\max_j P(Y = j|X)\right) = 0.133.$





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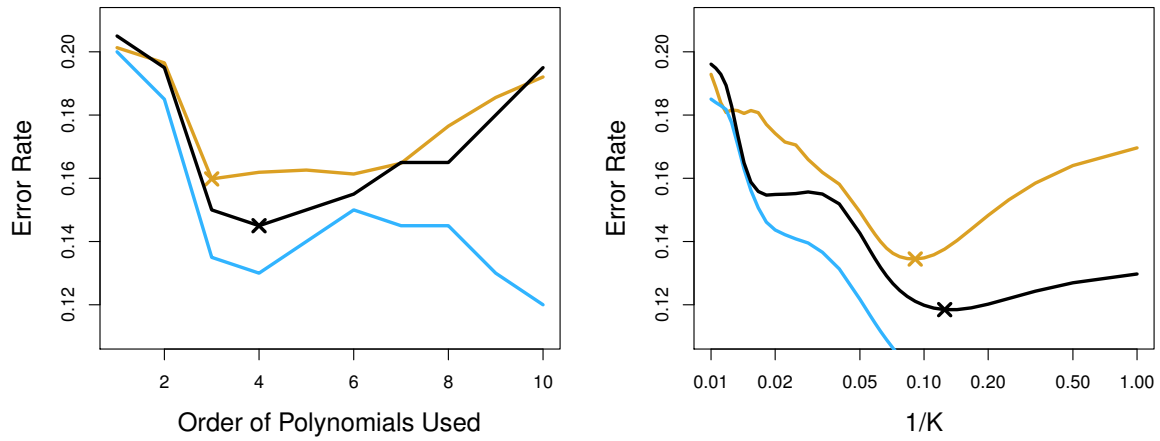
V –fold Cross-Validation versus KNN Classifier

- **Test error**, **training error** and **10-fold CV error** on the two-dimensional classification data.
- *Left:* Logistic Regression using polynomial functions of the predictors.
- *Right:* The KNN Classifier with different values of K (the number of neighbors).

✓ The 10 –fold CV error rate provides a pretty good approximation to the test error rate. While it somewhat underestimates the error rate, it reaches a minimum when fourth-order polynomials are used, which is very close to the minimum of the test curve.



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Bootstrap

The Bootstrap term comes from the saying *to pull oneself by one's own bootstrap.*

It consists in *sampling with replacement* a random sample from an observed sample of the same size as the observed sample.

Once the interest's statistics is computed on this random sample, this operation is repeated a (very large) number of times.

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Bootstrap Example

https://www.youtube.com/watch?v=X_UCeAGdAHE

- Suppose that we wish to invest a fixed sum of money in two financial assets that yield returns of X and Y, respectively, where X and Y are random quantities. We will invest a fraction α of our money in X and the remaining $1-\alpha$ in Y.
- Of course, we choose that α that minimizes the total risk of the investment.
- We want to minimize $Var(\alpha X + (1-\alpha)Y)$, yielding to

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

- Suppose we have 50 returns for both X and Y. Of course, we can get the point estimate of α.
- Let's simulate these returns with $\sigma_X^2 = 1$, $\sigma_Y^2 = 1.25$, $\sigma_{XY} = 0.5$.
- With these setting, we should obtain α=0.6.

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Boostrap sampling: how it works

Obs	X	Y
1	4.3	2.4
2	2.1	1.1
3	5.3	2.8

Original Data (Z)

Z^{*1}

Obs	X	Y
3	5.3	2.8
1	4.3	2.4
3	5.3	2.8

$\hat{\alpha}^{*1}$

Z^{*2}

Obs	X	Y
2	2.1	1.1
3	5.3	2.8
1	4.3	2.4

$\hat{\alpha}^{*2}$

Z^{*B}

Obs	X	Y
2	2.1	1.1
2	2.1	1.1
1	4.3	2.4

$\hat{\alpha}^{*B}$

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One sample-based estimate

```
> set.seed(10) #for reproducibility > s=cbind(c(1,0.5),c(0.5,1.25))
> C=mvrnorm(50,c(0,0),s)
> X=C[,1]
> Y=C[,2]
> SX=var(X)
> SY=var(Y)
> SXY=cov(X,Y)
> alpha=(SY-SXY)/(SX+SY-2*SXY)
> alpha
```

```
[1] 0.5961782
```

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How to get the boostrap estimate?

- We can proceed with a bootstrap approach with, say $B = 1000$ replications, in this way.
- Let $\mathcal{X}$ be the data set consisting in n pairs of observations X_i, Y_i
  - Randomly select n observations with replacement from the data set, call this new data set $\mathcal{X}_b$ with $b = 1, \dots, B$
  - Compute the quantity $\hat{\alpha}_b$ the bootstrap sample estimate of α
  - Repeat the operation B times, achieving $\hat{\alpha}_1$,…, $\hat{\alpha}_b$,…, $\hat{\alpha}_B$
- Compute the bootstrap estimate as

$$\hat{\alpha} = \frac{1}{B} \sum_{b=1}^{B} \hat{\alpha}_b$$

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What about the accuracy of our boostrap estimate?

- Compute the standard error of the bootstrap estimate as

$$\mathrm{SE}(\hat{\alpha}) = \sqrt{\frac{1}{B}\sum_{b=1}^{B}\left(\hat{\alpha}_b - \frac{1}{B}\sum_{b=1}^{B}\hat{\alpha}_b\right)^2}$$

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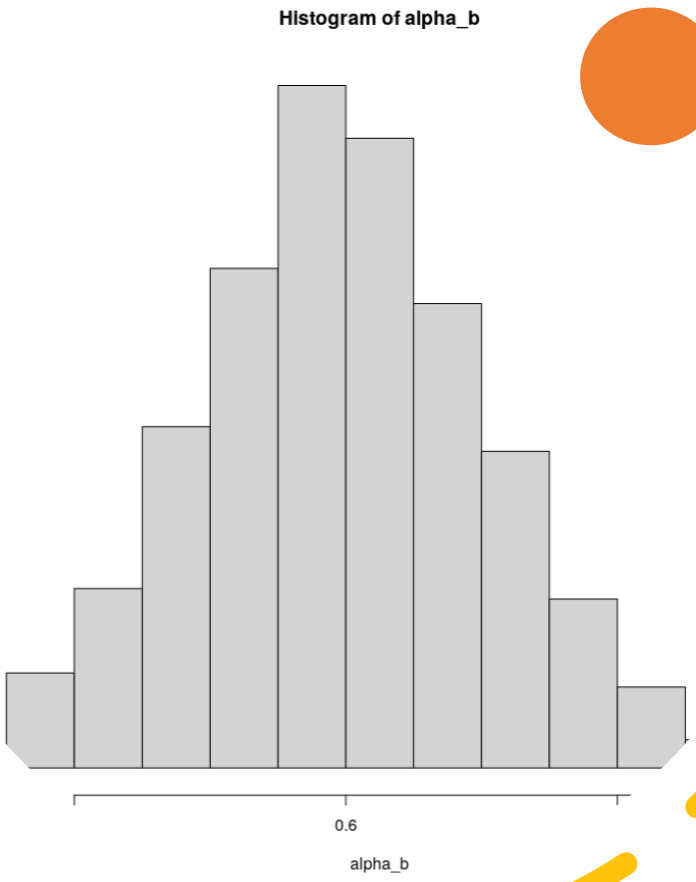
Bootstrap estimation

```
> B=1000

> alpha_b <- rep(0,B) # let initialize an empty vector of B values

> for (b in 1:B) { # a cycle of B bootstrap replications

> i <- sample.int(50, 50, replace = T) # resampling: 50 indexes from
1 to 50 with *replacement*

> x <- X[i] # i-th bootstrap sample X values

> y <- Y[i] # i-th bootstrap sample Y values

> # i-th alpha estimate
> SX=var(x)
> SY=var(y)
> SXY=cov(x,y)
> alpha_b[b] <- (SY-SXY)/(SX+SY-2*SXY)

> if (b<=4) plot(x,y) # draw a scatterplot of the first four bootstrap
samples
>}

> mean(alpha_b)
```



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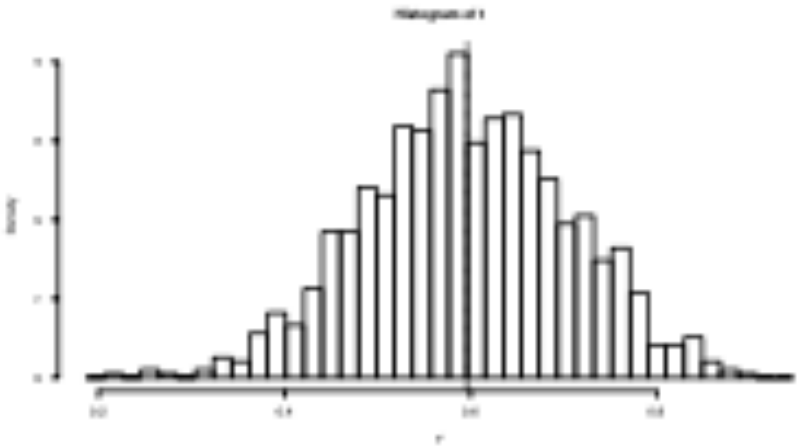
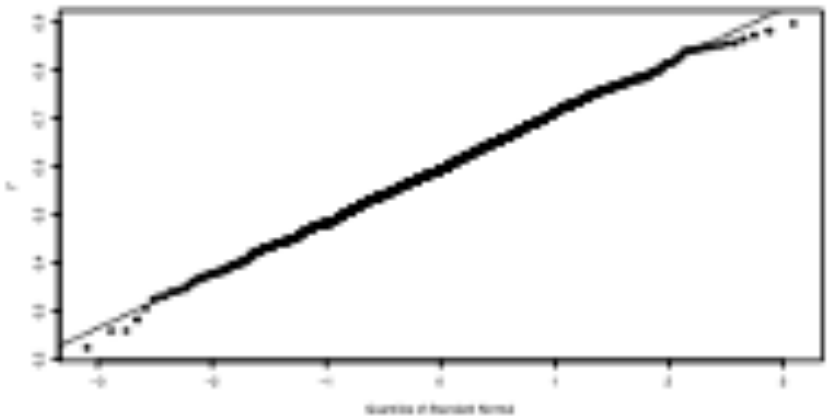
The results





alpha	bias	SE
0.5962	-0.0009	0.1105

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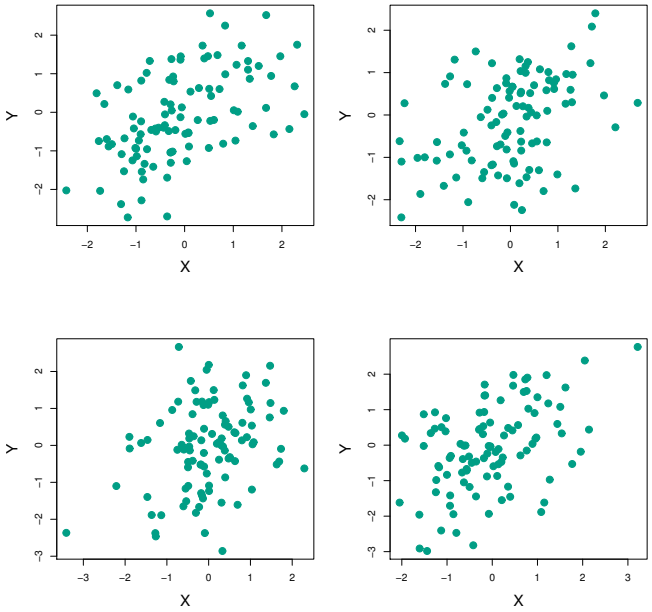
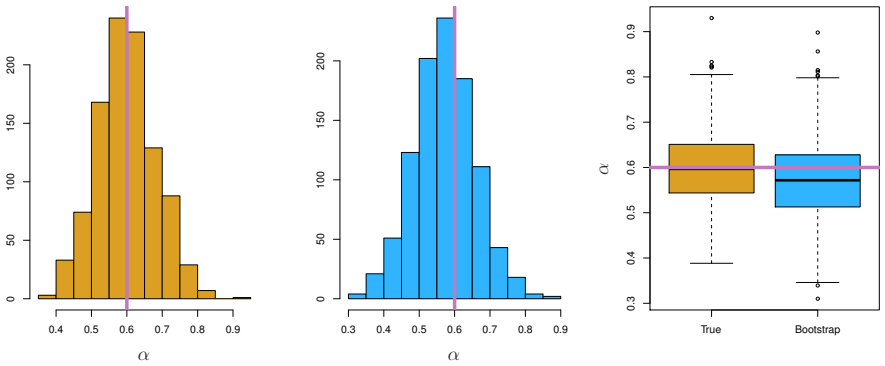


$\alpha = 0.576, 0.532, 0.657, 0.651$



Simulation setting:
100 simulated returns for investments X and Y and the resulting estimates for α

- Left: A histogram of the estimates of α obtained by generating 1,000 simulated data sets from the true population.
- Center: A histogram of the estimates of α obtained by generating 1,000 bootstrap samples from a single data set.
- Right: Boxplots of the estimates of α with the true value in the pink line.

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Purposes of bootstrap

- Bootstrap confidence intervals
  - Percentile bootstrap CI
  - Basic bootstrap CI
  - Studentized bootstrap CI
  - BCA bootstrap CI
- Estimation of the distribution of the sample statistics
- Accuracy of regression models
- A lot of other things....

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Bootstrap Confidence Interval

- Let $\hat{\theta}^{\mathcal{B}} = \hat{\theta}_1, \dots, \hat{\theta}_b, \dots, \hat{\theta}_B$ be the distribution of the statistic of interest through the bootstrap iterations.
- Let $\hat{\theta}$ be the point estimate in the sample.
- Let $\hat{\theta}^{\mathcal{B}}_{(1-\alpha/2)}$ be the $(1-\alpha)$-th percentile of the distribution of $\hat{\theta}^{\mathcal{B}}$.
- Let G be the (E)CDF of $\hat{\theta}$.
  - **Basic Bootstrap Confidence Interval** when the population model is symmetric but not known
    $$\left[2\hat{\theta} - \hat{\theta}^{\mathcal{B}}_{(1-\alpha/2)}; 2\hat{\theta} - \hat{\theta}^{\mathcal{B}}_{\alpha/2}\right]$$
  - **Percentile Bootstrap Confidence Interval** when there is no information on the population model distribution:
    $$\left[\hat{\theta}^{\mathcal{B}}_{\alpha/2}; \hat{\theta}^{\mathcal{B}}_{(1-\alpha/2)}\right]$$

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Studentized Bootstrap Confidence Interval

- For each bootstrap sample get

$$t_b = \frac{\hat{\theta}_b - \hat{\theta}}{SE(\hat{\theta}_b)}$$

where $SE(\hat{\theta}_b)$ denotes the standard error of the statistic within the b-th bootstrap replication.

- The $(1-\alpha)$ confidence interval is

$$\left[\hat{\theta} - t^{\mathcal{B}}_{(1-\alpha/2)} SE(\hat{\theta}); \hat{\theta} - t^{\mathcal{B}}_{(\alpha/2)} SE(\hat{\theta})\right]$$

where $SE(\hat{\theta})$ is the standard error of the estimate computed on the original data and $t^{\mathcal{B}}_{(1-\alpha/2)}$ is the $(1-\alpha/2)$-th percentile of the distribution of the quantity t along the B bootstrap replications.

- Drawback: double bootstrap in some cases.

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BCA Bootstrap Confidence Interval

- ***BCA (Bias Corrected and Accelerated Bootstrap)*** (Efron, 1987) adjusts for both bias and acceleration of the variance (skewness) in the bootstrap distribution, yielding reasonably narrow intervals.
- Suppose that for some monotone increasing transformation f, some bias constant z_0, and some acceleration constant a, the following relationship holds for $\phi = f(\theta)$:

$$\frac{\hat{\phi}-\phi}{\sigma} \sim N(-z_0, 1); \qquad \sigma = 1 + a\phi$$

- If the conditions above are respected, then the BCA confidence interval is

$$\left[\hat{G}^{-1}\left(\phi\left(z(\alpha)\right)\right); \hat{G}^{-1}\left(\phi\left(z(1-\alpha)\right)\right)\right],$$

where $z(\alpha) = z_0 + \frac{z_0 + z^{(\alpha)}}{1-\alpha\left(z_0+z^{(\alpha)}\right)}$

- In order to construct the BCA interval, one has to estimate z_0 and a. It is not necessary to know $f(\theta)$.

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